



DATA WRANGLING AND RETRIEVAL

School Type & Academic Performance in Catalonia

Insights from Four Years of PAU and Enrolment Data
(2020/2021 – 2023/2024)

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An aerial photograph of a city, likely Santiago, Chile, with a large, light-colored classical building with columns in the foreground. The city is densely packed with buildings, and mountains are visible in the background under a clear sky. The image is used as a background for a presentation slide.

01

INTRODUCTION

Research topic, objectives, and data sources

| Research Topic & Objectives

TOPIC

This study investigates the relationship between **school type** (public vs. private) and **student academic performance** in **all regions of Catalonia**, measured by average university entrance examination (**PAU**) scores over four academic years (2020/2021 – 2023/2024).

OBJECTIVES

1

Compare PAU Scores

Compare average PAU scores between public and private schools across all regions of Catalonia

2

Analyse Score Trends

Analyse performance trends over 4 academic years to identify patterns of change over time

3

Explore Enrolment Link

Explore whether the number of student enrolments is associated with academic achievement



Regions = Territorial education services. They are administrative education districts used by the Catalan Department of Education.

| Data Sources



DF1 — PAU Scores

2020 – 2024

Official PAU scores aggregated by region and school type. Contains average grade data for public and private schools across all Catalan regions.



DF2 — Student Enrolment Data

2020 – 2024

Student enrolment records by region and school type, obtained partially through **web scraping**.

Comprised of 8 separate data files that were individually extracted, cleaned, and concatenated.



Both datasets were retrieved from official Catalan government sources.
Enrolment data includes secondary and high school.

02

METHODOLOGY

Data wrangling workflow and project management

| Data Cleaning Workflow



KEY PROCESS STEPS

- ✓ **9 independent datasets** were cleaned individually,
- ✓ **Pre-concatenation validations** – validate the datasets before concatenating to ensure data integrity.
- ✓ **Analysis** of relationships between school type, region, enrolment size, and academic performance.

| Project Management & Best Practices

CHALLENGES

Multiple Data Sources

9 independent datasets had to be cleaned individually before any concatenation could take place.

Inconsistent Regional Names

Regional names varied across sources, requiring **mapping** dictionary for unification.

Validation Complexity

Extensive **validations** were required before combining to prevent data loss and misalignment.

BEST PRACTICES

Track DataFrame Versions

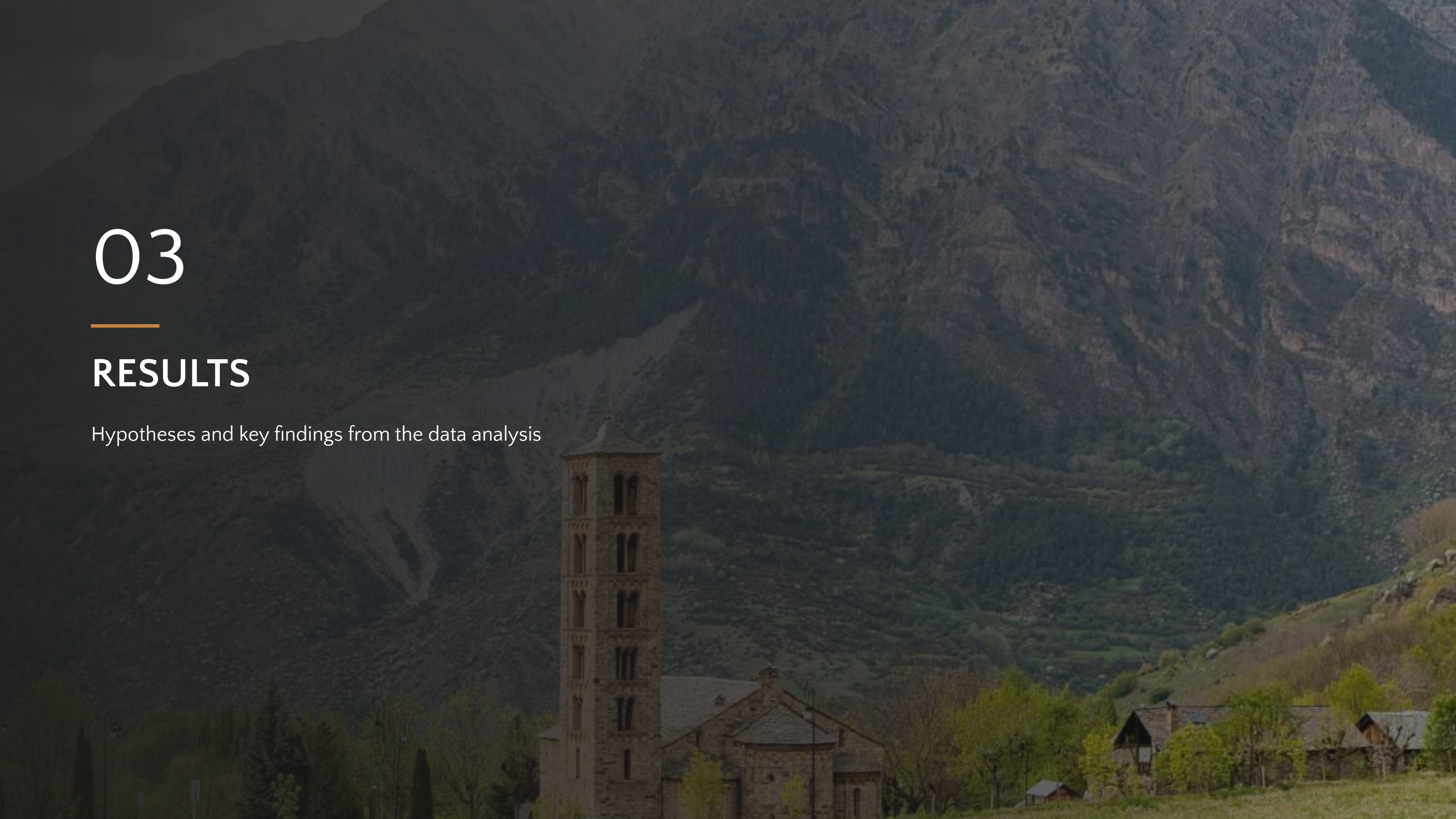
Maintained detailed notes of all dataframe names and versions throughout the process.

Test Before Overwriting

Tested all modifications (concat, merge, groupby, drop) before reassigning to variables.

Document Assumptions

Recorded all assumptions, hypotheses, and insights during the cleaning process.

The background of the slide is a photograph of a mountainous landscape. In the foreground, a stone church tower with multiple arched windows stands prominently. To its right, there are some smaller buildings and trees. The background is dominated by steep, rocky mountains with patches of green vegetation. The overall tone is somewhat muted and atmospheric.

03

RESULTS

Hypotheses and key findings from the data analysis

| Hypotheses

HYPOTHESIS 1

Private school = Higher performance

Students attending **private schools** are expected to obtain higher average PAU scores compared to students in public schools.

HYPOTHESIS 2

Enrolment & Competitiveness

Regions with the **highest number of students** would have higher PAU scores due to increased competitiveness.

ANALYSIS APPROACH



Data
Wrangling



Descriptive
Statistics

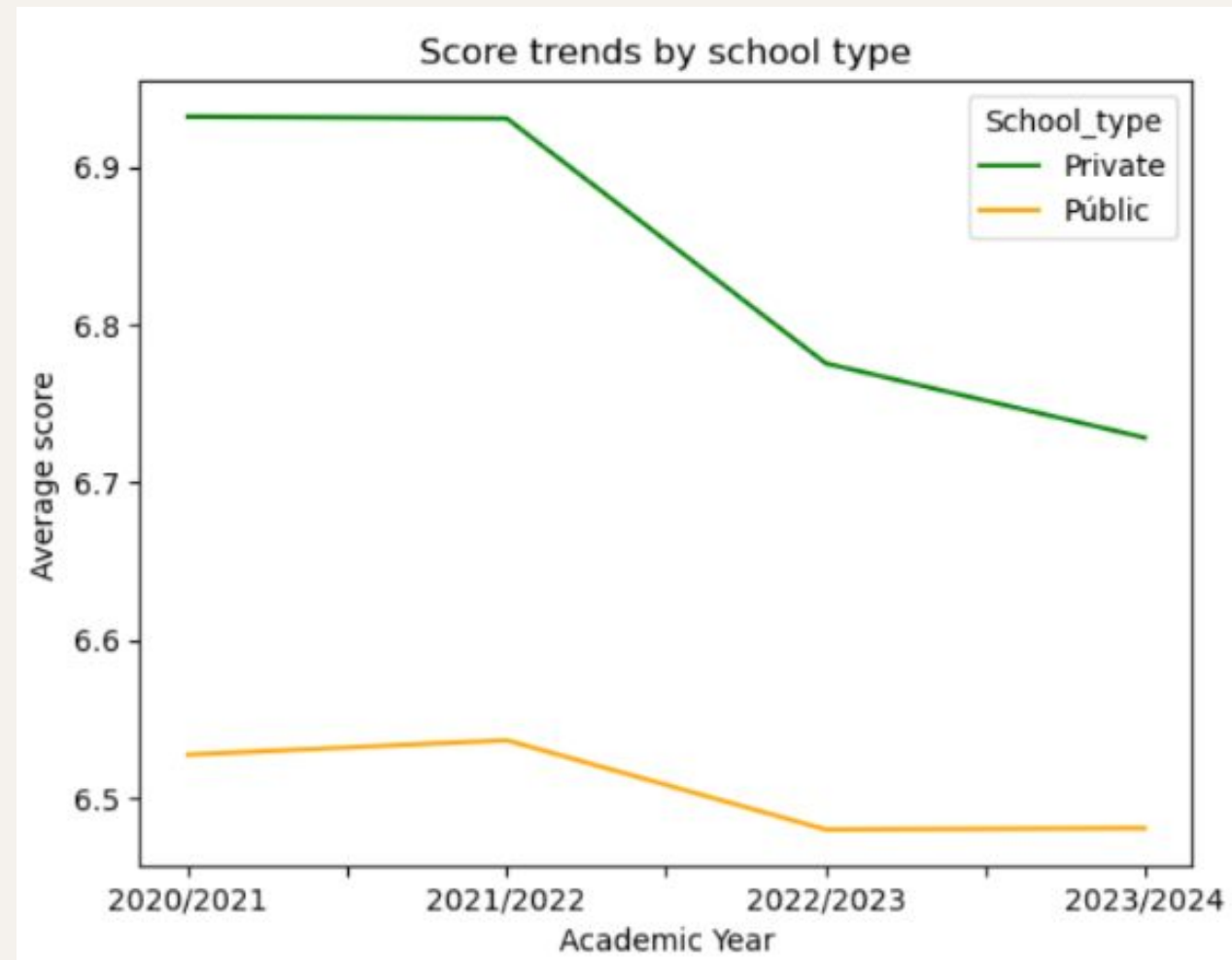


Trend
Analysis



Comparative
Analysis

| Average PAU Scores by School Type

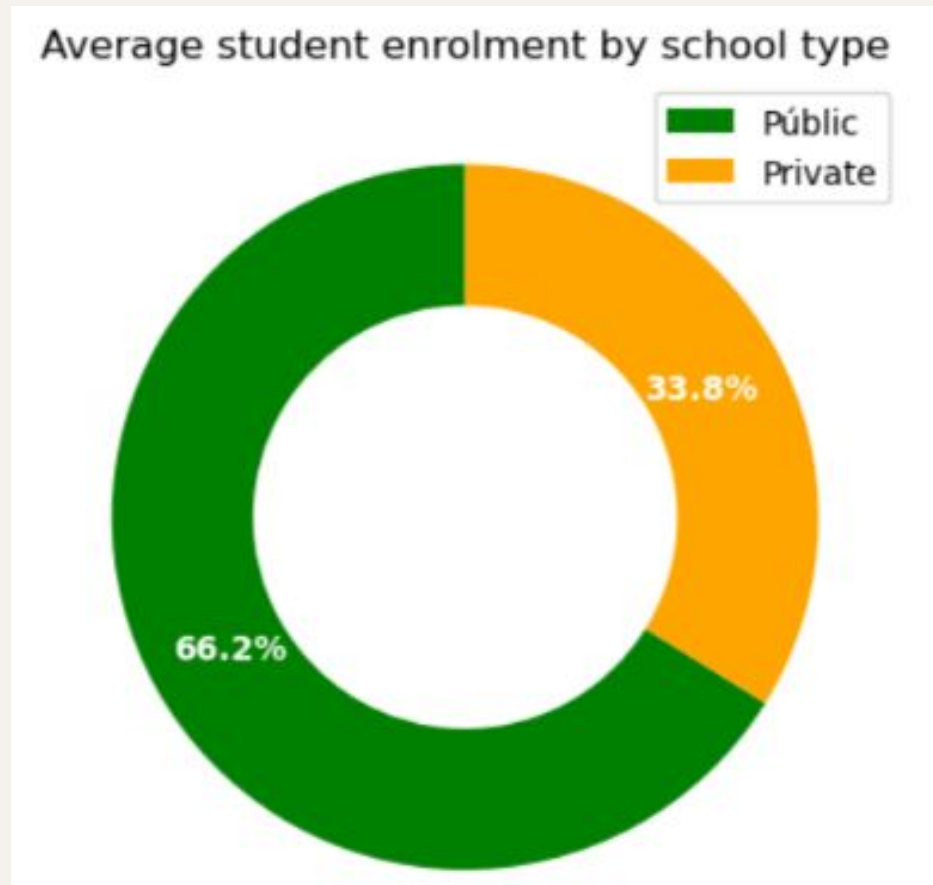


KEY FINDINGS

- **Private schools** consistently achieved higher average PAU scores across all 4 years
- Both school types show a **downward trend** in average scores over time
- The decrease appears **larger for private schools**, though they remain ahead

Conclusion: Hypothesis 1 is supported — private schools consistently outperform public schools in PAU scores.

| Enrolment Distribution by School Type



REGIONAL ENROLMENT STATS

HIGHEST ENROLMENT

Barcelonès Private — 11.570 students (avg)

Barcelonès Public — 9.116 students (avg)

LOWEST ENROLMENT

Alt Pirineu i Aran Private — 789 students (avg)

Terres de l'Ebre Public — 2.007 students (avg)

Public schools enrolled roughly **twice as many (51%) students** as private schools across Catalonia. Barcelonès dominates enrolment figures, while rural regions like Alt Pirineu and Terres de l'Ebre have significantly smaller student populations.

| Enrolment size vs. Academic Performance

HIGHEST SCORE

Maresme – Vallès Oriental

Private schools — highest avg. PAU score over 4 years

Enrolment size rank: **11th out of 21 regions**

LOWEST SCORE

Barcelonès Public

Public schools — lowest avg. PAU score over 4 years

Enrolment size rank: **2nd highest** in Catalonia



- ✗ **Hypothesis 2 is not supported.** Enrolment size is not a strong predictor of academic performance. The highest-scoring region ranks only 11th in enrolment.

An aerial photograph of a lush, forested valley. The trees are in various shades of green and yellow, suggesting autumn. In the center-left, a small building with a conical roof is visible. To its right, a small pond or clearing is surrounded by trees. The background shows rolling hills covered in dense forest.

04

NEXT STEPS

Limitations, and future work

| Limitations & Future Work

LIMITATIONS



No Statistical Testing

Statistical significance was not tested. Future work should include formal hypothesis testing to validate observed differences.



No Socioeconomic Data

Socioeconomic status was not considered — a significant confounding variable that may explain performance differences.



School Distribution

Proportion of school types by region was not analysed. High-scoring regions may have more educational resources.

FUTURE DIRECTIONS

- Conduct **formal statistical tests** to determine whether observed differences between school types are statistically significant.
- Incorporate **socioeconomic indicators** (family income, parental education) to control for confounding variables in performance analysis.
- Analyse the **proportion of school types by region** to examine whether higher-scoring regions have greater access to educational resources.

DATA WRANGLING AND RETRIEVAL

Thank You

School Type & Academic Performance in Catalonia

Questions & Discussion

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